Reasoning-augmented Vision-Language Models for Improved Competency in Scene Text Recognition

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***Abstract*—This study investigates the use of Vision-Language Models (VLMs) for Korean scene text recognition, with a focus on enhancing their reasoning capability for surveillance applications. We address the challenge of Korean banner understanding from surveillance camera feeds, which requires both accurate text recognition and classification into political, public, or private categories. We curated a dataset of Korean banner images including warped and synthetic flat images, and trained the Qwen2.5-VL-3B model as our backbone. To strengthen reasoning, we constructed reasoning-augmented training data that pairs extracted text with intermediate rationales, and designed inference prompts that encourage step-by-step explanations before classification. Experimental results show that incorporating explicit reasoning substantially improves both recognition and classification performance compared to baselines. Reasoning prompts alone provide significant improvements in classification accuracy, while the combination of reasoning-augmented training and reasoning prompts achieves the best overall performance. These findings highlight the importance of reasoning supervision and prompting for building more interpretable VLMs in real-world text understanding tasks.**

***Index Terms*—Vision-Language Model (VLM), Korean Scene Text Recognition, Banner understanding, Chain-of-thought**

I. Introduction

In recent years, the rapid development of smart city infrastructure has highlighted the importance of surveillance technologies for ensuring public safety, efficient urban management, and intelligent transportation systems [1]–[4]. Within this context, the ability to automatically interpret textual information embedded in the environment, such as street signs, commercial banners, license plates, and public notices, has become increasingly critical. Scene Text Recognition (STR) plays a pivotal role in enabling machines to understand such unstructured visual information, thereby supporting applications ranging from traffic monitoring and regulatory compliance to the detection of illegal advertisements and public safety alerts. In this work, we specifically focus on Korean banner understanding from surveillance camera feeds, which involves both recognizing the textual content and classifying banners into categories such as political, public, or commercial for effective monitoring and regulation.

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While significant progress has been made in scene text recognition through both traditional text recognition pipelines and deep learning-based methods [5]–[8], the unique challenges of real-world scenarios, including diverse fonts, cluttered backgrounds, and low-quality imaging conditions, continue to drive research demand. These challenges are particularly pronounced in multilingual contexts such as Korean, where character complexity and irregular layouts increase the recognition difficulty.

To address these challenges, Vision-Language Models (VLMs) have emerged as a powerful alternative, offering an end-to-end approach that integrates both visual and linguistic information for robust scene text understanding. While current state-of-the-art VLMs have shown remarkable success in document analysis and general image reasoning tasks [9], they are predominantly trained on English and Chinese datasets, limiting their effectiveness for languages with different characteristics. This creates a need for targeted approaches to enhance their capabilities for specific domains and languages.

In this study, we enhance the reasoning capability of Vision-Language Models (VLMs) for Korean scene text understanding. We hypothesize that a model's ability to provide a logical, step-by-step rationale for its classification decisions is crucial for achieving robust and reliable performance. By guiding the VLM to explicitly generate reasoning traces, we aim to improve logical consistency and accuracy beyond simple character recognition. Our training and inference strategies are designed to systematically evaluate the impact of reasoning supervision and reasoning-based prompting on the model's performance.

The main contributions of this paper are summarized as follows:

- We demonstrate that reasoning-based prompts substantially improve classification accuracy in Korean scene text tasks, even without reasoning supervision during training.
- We quantify the additional performance gains obtained when a VLM is explicitly trained on a reasoning-augmented dataset that includes intermediate rationales alongside final labels.
- We propose a framework that combines reasoning supervision and reasoning-based prompting to classify Korean

banners while providing interpretable justifications for improved transparency.

II. Related Works

A. Traditional Approaches to Scene Text Recognition

For decades, the field of Scene Text Recognition (STR) has been dominated by multi-stage pipelines that perform text detection, segmentation, and recognition in sequence [5]–[8]. These methods, including models like CRNN [10] and Transformer-based architectures, work well for structured documents with uniform layouts and clear backgrounds. However, they face challenges in real-world scenes with diverse fonts, complex backgrounds, and geometric distortions. This is especially difficult for languages with complex characters like Korean, where the structure of Hangul characters and irregular layouts on banners can reduce recognition accuracy. This sequential approach processes each text element separately, often missing the broader visual and linguistic context.

B. Vision-Language Models for Integrated Text Analysis

Recently, Vision-Language Models (VLMs) [11]–[13] have emerged as a strong alternative, offering an end-to-end approach that processes both visual and linguistic information simultaneously. This integrated approach allows these models to understand the entire scene, including a banner's overall layout and the relationships between different text elements. By learning connections between visual patterns and text meaning, VLMs can achieve better recognition accuracy, even when text is partially hidden or distorted by complex graphics and logos. Unlike traditional methods that process individual characters, VLMs can use surrounding context to distinguish visually similar characters and correctly interpret spacing. However, many existing VLMs are primarily trained on English and Chinese datasets, which limits their performance on languages like Korean.

C. Chain-of-Thought (CoT) Reasoning

The Chain-of-Thought (CoT) technique [14]–[16] is a method used to encourage large language models (LLMs) to break down complex problems into explicit, intermediate reasoning steps before reaching a final conclusion. This approach improves the logical consistency and accuracy of a model's output. For tasks like banner classification, CoT enables the model to go beyond simple keyword matching and instead analyze the text's theme, expressions, and context in a step-by-step manner. While CoT has been widely applied in general LLM applications, its use in improving the reasoning capabilities of VLMs for specific, real-world tasks like scene text classification remains less explored.

III. Methodology

This section describes our methodology to enhance the reasoning competency of a Vision-Language Model (VLM) for Korean scene text recognition. We first explain how the dataset was prepared, then describe the choice of base model and our training strategy. Finally, we introduce reasoning-augmented training data and reasoning-based prompt design for inference.

A. Data Preparation

A straightforward way to construct ground-truth (GT) labels is to use a text detector, run a conventional text recognition engine, and combine the recognized words. However, such a simple approach often produces inaccurate bounding boxes, disordered sequences, and unnatural reading flow, which reduces the quality of supervision.

To address these issues, we re-labeled an existing dataset of Korean banner images (both flattened design drafts and real-world photos). Using the GPT-4.1 API [17], we generated high-quality GT text that better reflects the logical order and layout of banners. Instead of explicitly encoding line breaks, we organized the text into semantically consistent and well-separated units, allowing the model to learn natural reading flow more effectively.

Although VLMs can interpret global textual information, they often struggle with localized details. We adopted a Grounding DINO model [18] pre-trained with text supervision and fine-tuned it for our domain to provide bounding boxes [19]. We also applied geometric warping to create flattened banner images, using the VLM's strong performance on document-style inputs. Additionally, synthetic flat banner images from design drafts were added to increase stylistic and linguistic diversity. Finally, we applied a verification stage using PaddleOCR [8] to filter out noisy annotations, creating a cleaner and more reliable dataset.

B. Model Selection and Fine-tuning

For our VLM backbone, we selected Qwen2.5-VL-3B [13]. Training a vision-language model entirely from scratch would require extensive computational resources and a very large-scale corpus, which are beyond the scope of our research setting. Unlike models such as GOT-OCR [20], PerceptionLM [21], and SmolVLM [22], which are primarily optimized for general scene understanding but show low accuracy in recognizing Korean text, Qwen2.5-VL was pre-trained on a large multilingual corpus that includes Korean. This pre-training makes it more suitable for handling the complex character structures and irregular layouts commonly found in Korean banners. We trained only the vision encoder while keeping the language model frozen.

C. Reasoning-augmented Training Data

To further improve reasoning, we constructed conversational training data inspired by LLaVA-CoT [23]. Each example required the model to (1) extract text from a banner, (2) explain the reasoning behind the classification, and (3) output the final class (political, public, or private). We prepared two variants:

- **Reasoning-augmented dataset**: includes intermediate rationale along with the final label.
- **Answer-only dataset**: contains only the final label.

This design allows us to isolate the effect of reasoning supervision during training, independent of prompt design at inference time (cf. Table I).

Since creating reasoning traces is expensive, we used a semi-automatic pipeline for the reasoning-augmented dataset.

TABLE I: Examples of reasoning-augmented and answer-only training data. Korean text is shown with English translation in parentheses.

Type	Example Conversation (simplified)
Reasoning-augmented	Human: <image>현수막에서 보이는 텍스트를 추출해줘. (Human: Extract the text from the banner.) GPT: ”노원역, 900만원대 아파트, 초역세권, 032-870-9788” (GPT: ”Nowon station, apartment around 9M KRW, near station, 032-870-9788”) **Human: 이 텍스트를 바탕으로 정당, 공공, 민간 중 어떤 유형인지 판단한 근거를 설명해줘.** **(Human: Based on this text, explain the reasoning for classification.)** **GPT: ”부동산 상품을 홍보하는 내용으로 보이므로 민간 현수막으로 분류 가능.”** **(GPT: ”Since it promotes a real estate property, it is likely a private banner.”**) Human: ”최종적으로 이 현수막은 정당, 공공, 민간 중 어디에 해당하나요?” (Human: Finally, which category does this banner belong to: political, public, or private?) GPT: ”민간” (GPT: ”Private”)
Answer-only	Human: <image>현수막에서 보이는 텍스트를 추출해줘. (Human: Extract the text from the banner.) GPT: ”노원역, 900만원대 아파트, 초역세권, 032-870-9788” (GPT: ”Nowon station, apartment around 9M KRW, near station, 032-870-9788”) Human: ”최종적으로 이 현수막은 정당, 공공, 민간 중 어디에 해당하나요?” (Human: Finally, which category does this banner belong to: political, public, or private?) GPT: ”민간” (GPT: ”Private”)

We first generated initial rationales using Kanana-1.5-3b-instruct [24] and VARCO-Vision-14B [12], both trained on large-scale Korean data, and then performed light human post-editing to correct major errors and ensure consistency with our schema. This procedure significantly reduced annotation cost while maintaining the quality of reasoning supervision.

D. Prompt Design for Inference

At inference, we evaluated the effect of reasoning through two prompt designs:

- **Direct Prompt**: instructs the model to extract text and directly output the classification.
- **Reasoning Prompt**: requires the model to first extract text, then justify its decision step by step before giving the final class.

The reasoning prompt was carefully designed with explicit classification criteria (e.g., political banners contain party names or candidates, public banners include government or institution notices, and private banners consist of commercial ads). By requiring justification, the reasoning prompt produces more interpretable and reliable outputs. Examples are shown in Table II.

IV. Experiments and Results

This section presents our experimental setup and the results that demonstrate the effectiveness of reasoning-augmented training and reasoning-based prompting. The primary objective is to measure the impact of reasoning supervision on both text recognition and banner classification in Korean surveillance scenarios.

A. Dataset and Training Setup

As described in Section III-A, we used a task-specific detector [19] based on Grounding DINO [18] to provide cropped banner regions as input. The training set was composed of three equally sized sources: $2,800$ cropped images from real surveillance scenes, $2,800$ warped images that simulate flattened documents to use the VLM's strength in structured text understanding, and $2,800$ synthetic flat banner images resembling design drafts. In total, 8,400 samples were used for training, while 1,024 images were reserved for validation.

Training was conducted on the Qwen2.5-VL-3B model using a single NVIDIA A100 GPU. Only the vision encoder was updated, while the language model remained frozen. The model was trained for 20 epochs with mixed-precision optimization (bfloat16). To ensure stable batch behavior, the input resolution was fixed at 451,584 pixels. Optimization used a linear learning rate schedule with a warm-up phase (10% of training steps, capped at 1,000 warm-up steps), weight decay of 0.01 for regularization, and gradient clipping with a maximum norm of 1.

B. Evaluation Metrics

Evaluation covered both text recognition quality and downstream classification accuracy.

For recognition, we report Character Error Rate (CER) and Word Error Rate (WER), computed as the normalized Levenshtein distance between the ground truth and predicted sequences. Specifically, if S, D, and I denote the numbers of substitutions, deletions, and insertions required to align a hypothesis h to a reference r of length N, the error rate is defined as

$$\text{Error Rate} = \frac{S + D + I}{N}. \tag{1}$$

Lower CER and WER indicate better recognition performance.

To capture semantic similarity beyond strict sequence matching, we also report chrF (character n-gram F-score) and ROUGE-L (longest common subsequence based recall/precision). Additionally, we compute a token-level F1

TABLE II: Examples of Direct and Reasoning prompts. Korean instructions are shown with English translation in parentheses.

Type	Example Prompt
Direct Prompt	”이 현수막에 적힌 글자를 읽어주세요. (Please read the text written on this banner.) 출력 형식: (Output format:) - 현수막 내용: [읽은 텍스트] (Banner text: [recognized text]) - 분류: [정당/민간/공공 현수막] (Classification: [political/public/private banner])”
Reasoning Prompt	”위 현수막의 텍스트를 꼼꼼하게 읽고, 내용을 말해주세요. (Carefully read the text in the banner and describe it.) 그 후 이 현수막이 ’정당 현수막’, ’민간 현수막’, ’공공 현수막’ 중 어디에 해당하는지도 함께 판단해주세요. (Then decide whether the banner belongs to political, private, or public.) 판단 기준: (Criteria:) 1. 정당 현수막: 정당명, 후보자, 선거 관련 문구 포함 (Political: contains party names, candidates, or election slogans) 2. 민간 현수막: 상업 광고, 학원/상점/개인 목적 문구 포함 (Private: contains commercial ads, private institutes, or individual promotion) 3. 공공 현수막: 정부, 지자체, 공공기관 명칭 또는 정책, 단속, 행정 안내 등 (Public: contains government, municipality, or public service information) 출력 형식: (Output format:) - 현수막 내용: 현수막 내용 읽기 (Banner text: recognized content) - 분류 결과: 정당 현수막 / 민간 현수막 / 공공 현수막 (Classification: political / private / public banner) - 판단 이유: 텍스트나 문맥을 근거로 분류 이유 설명 (Reasoning: explanation based on text or context)”

score, which measures overlap between sets of extracted tokens regardless of order, providing a more layout-independent perspective. For chrF and ROUGE-L, higher scores indicate better semantic alignment with the ground truth, while Token F1 measures how well the model captures individual meaningful units from the text.

For the final task, we evaluate classification accuracy across three categories (political, public, and private), which reflects the reasoning ability of the model beyond basic text recognition performance.

C. Experimental Design

To evaluate the effectiveness of our approach, we compared different combinations of training supervision and inference prompting. We first tested models trained only with text recognition and label supervision, without reasoning traces. For these models, we evaluated both direct prompting and reasoning-based prompting at inference. We then introduced reasoning-augmented training, where the dataset included intermediate rationales in addition to the final label, and evaluated the model with reasoning prompts.

As shown in Table III, this design allows us to isolate three factors: (1) the benefit of reasoning prompts at inference, (2) the effect of training with Korean banner data, and (3) the additional impact of reasoning-augmented training.

D. Effect of Reasoning Prompts at Inference

Applying reasoning prompts to the baseline Qwen model led to a significant improvement in classification accuracy, from 28.46% with direct prompting to 65.90%. A similar trend was observed for the trained model, where accuracy rose from 24.64% to 65.62%. These results confirm that reasoning prompts alone can substantially enhance the decision-making ability of a VLM, encouraging the model to generate intermediate explanations that guide more accurate classification. Interestingly, while recognition metrics such as CER and WER improved only slightly, content-based metrics (chrF, ROUGE-L, Token F1) showed more noticeable improvements, reflecting the model's improved semantic alignment.

E. Effect of Training with Korean Data

Training on our curated Korean banner dataset yielded clear gains in recognition quality compared to the pre-trained Qwen baseline. With direct prompting, CER decreased from 0.6546 to 0.5932 and chrF improved from 0.3795 to 0.5330. These improvements demonstrate that domain adaptation via training enables the model to better capture the visual and linguistic characteristics of Korean banners. However, classification accuracy remained low under direct prompting (24.64%), indicating that reasoning at inference is still critical for robust decision-making.

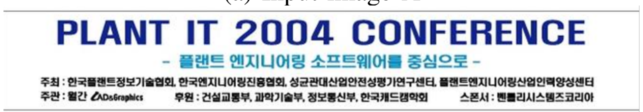
F. Effect of Reasoning-augmented Training

Finally, when reasoning-augmented training was combined with reasoning prompts, the model achieved the best overall performance, reaching 70.96% classification accuracy. Recognition metrics also improved, with CER reduced to 0.5226 and chrF increased to 0.5931, the highest among all settings. This

TABLE III: Performance comparison on Korean banner recognition and classification. Qwen refers to the pre-trained baseline model, and Ours is trained on the Korean dataset. Arrows (↓/↑) indicate whether lower or higher values are better.

Method	CER (↓)	WER (↓)	chrF (↑)	ROUGE-L (↑)	Token F1 (↑)	Class. Acc. (↑)
Qwen-2.5-VL (baseline) + Direct prompt	0.6546	0.7794	0.3795	0.4257	0.4587	28.46
Qwen-2.5-VL (baseline) + Reasoning prompt	0.6219	0.7852	0.4490	0.4875	0.5242	65.90
Ours (trained) + Direct prompt	0.5932	0.7696	0.5330	0.5706	0.5938	24.64
Ours (trained) + Reasoning prompt	0.5508	0.7047	0.5652	0.6072	0.6315	65.62
Ours (trained, reasoning-aug.training) + Reasoning prompt	**0.5226**	**0.7041**	**0.5931**	**0.6294**	**0.6552**	**70.96**

(a) Input image A



(b) Input image B

(c) Input image C

Fig. 1: Examples of Korean advertisement text used for OCR evaluation.

TABLE IV: Qualitative results for Images A, B, and C. Symbols: misrecognition, ◆ missing.

Image	Method	OCR Result (Korean)
A	GT	위치갑 송도 1공구 아파트 및 단지 내 상가 포스코 자사고 앞 파격 조건 분양 032-870-6940
	Baseline	송도 1공구 아파트 및 단지 내 상가 포스코 자사고 앞 파격 조건 분양 ◆
	Proposed	위치갑 송도 1공구 아파트 및 단지 내 상가 포스코 자사고 앞 파격 조건 분양 032-870-6940
B	GT	PLANT IT 2004 CONFERENCE – 플랜트 엔지니어링 소프트웨어를 중심으로 – 주최: 한국플랜트정보기술협회, 한국엔지니어링진흥협회, 성균관대산업안전성평가연구센터, 플랜트엔지니어링산업인력양성센터 주관: 월간 CAD&Graphics 후원: 건설교통부, 과학기술부, 정보통신부, 한국캐드캠학회 스폰서: 벤틀리시스템즈코리아
	Baseline	PLANT IT 2004 CONFERENCE – 플랜트 엔지니어링 소프트웨어를 중심으로 – ◆
	Proposed	PLANT IT 2004 CONFERENCE, 플랜트 엔지니어링 소프트웨어를 중심으로, 주최: 한국플랜트정보기술협회, 한국엔지니어링진흥협회, 성균관대산업전성평가연구소, ◆, 주관: 월간 **AsGraphics**, 후원: 건설교통부, 과학기술부, 정보통신부, 한국캐드**대학교**, 스пон서: 벤**물리시웰즈**코리아
C	GT	(4행정) 30만원 → 15만원 방문수령 조건 (VAT 별도) 기간: 8월 20부터 ~ 소진시까지 배낭예초기 (30대 한정) 반값 세일 문의: 본사: 한국농기계(주), 041-751-0401, 금성면 금성공단로 19-18.
	Baseline	배낭예초기 (30대 향정) 반값세일 ◆
	Proposed	**이해결제(소집)** – 15만원 방문수령 조건 (VAT **불가**) 기간: 8월 20부터 ~ 소집시 배당예초기 (30대 한정) 반값세일 문의 본사: 국농기계(주), 041-751-0401, 금성면 금성공단로 19-18

shows that exposing the model to explicit reasoning traces during training strengthens its ability to structure and justify predictions, resulting in consistent improvements across both recognition and classification tasks.

G. *Qualitative Analysis*

Table IV presents qualitative comparisons between the baseline and our proposed method. The baseline model generally captures only large and prominent characters, producing partial outputs that miss finer details such as contact numbers, sponsors, or conditional statements. While this often yields clean results for main titles, the extracted text remains incomplete and fails to convey the full message of the banner. In contrast, the proposed method successfully recovers a much larger portion of the textual content, including smaller and denser phrases, thereby providing a more comprehensive representation of the banners. Nevertheless, errors still occur, particularly in rare proper nouns or noisy text regions, which indicates that while coverage has improved, recognition accuracy for challenging segments remains an open issue.

V. Limitations and Future Works

Although our study demonstrates the effectiveness of reasoning supervision and reasoning-based prompting for Korean banner understanding, several limitations remain. The size of our dataset (8,400 samples) is relatively small compared to the scale usually required for robust multimodal reasoning, and recognition errors are still observed in rare proper nouns or heavily degraded text. These limitations highlight the need for broader coverage and stronger noise-resistant training.

For future work, we plan to extend the dataset with large-scale synthetic banners where diverse fonts, layouts, and noise conditions can be systematically generated with reliable ground truth. Such expansion will allow the model to better generalize to real-world surveillance scenarios. Additionally, we aim to incorporate domain knowledge such as banner-specific terminology and common phrases, and explore more noise-resistant pretraining strategies to further enhance recognition accuracy and reasoning robustness.

VI. Conclusion

This study addresses the challenge of Korean banner understanding in surveillance scenarios by enhancing Vision-Language Models with reasoning capabilities. We investigated the impact of reasoning supervision during training and reasoning-based prompting during inference on both text recognition and classification performance. Our results show that reasoning prompts substantially improve classification accuracy, and combining them with domain-specific training and reasoning-augmented datasets yields the best performance. The proposed framework enables models to move beyond character-level recognition toward contextual understanding of banner content. This approach supports surveillance applications that require categorization of banners for regulatory compliance and monitoring. While recognition errors persist in challenging cases, the method offers improved interpretability compared to traditional approaches.

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